The Free-Time Gap over the Life Course: Gender, Family Stage, and Time Poverty in India

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Abstract

Free time is a basic dimension of well-being, yet gender gaps in it are usually reported as a single national average. Using every diary in India's Time Use Survey 2024 (533,719 persons), I trace the women-minus-men gap in free time, and in time poverty, across seven family stages from youth to old age. Women aged 15 and above have 15 minutes less free time a day than men (SE 0.8) and do 30 minutes more total paid and unpaid work (SE 1.1). The gap is widest among persons "young, never married (15-24)". A Kitagawa decomposition splits the women-minus-men gap of -15.1 minutes into -11.6 minutes arising within stages and -3.5 minutes arising from the different distribution of men and women across stages. Women are 2.1 percentage points more likely than men to be time-poor at a line of 60% of median free time, and 2.6 points more at 50%. Conditional on covariates, the gap varies more across family stages than across education, consumption, caste, religion or sector. I also report the gap with religious practice excluded from free time, a joint time and consumption poverty measure, the free-time difference between pensioners and workers aged 55 to 70, and the time cost of commuting for employed men and women.

Keywords: free time, leisure, time poverty, life course, gender, retirement, commuting, India
JEL codes: D13, I31, I32, J16, J22

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1 Introduction

Time is a resource, and free time is one of the clearest measures of well-being that a time budget yields. A person with little discretionary time is constrained even when household income is adequate, because the hours needed to earn, cook, clean and care leave no room for rest, social life or civic participation (Vickery 1977; Goodin et al. 2008). The Commission on the Measurement of Economic Performance and Social Progress recommended that leisure and the distribution of time enter the measurement of social progress alongside income (Stiglitz et al. 2009), and time-use surveys are the instrument that makes this possible (Juster and Stafford 1991; Krueger et al. 2009).

In high-income countries, a large literature has documented how leisure is distributed. Leisure has risen over the long run but has diverged by education (Aguiar and Hurst 2007; Ramey and Francis 2009; Sevilla et al. 2012). Women's free time is shorter than men's and is more often fragmented or combined with childcare (Bittman and Wajcman 2000; Mattingly and Bianchi 2003), and the gender gap in free time persisted as paid and unpaid work converged (Sayer 2005). In developing countries the evidence is thinner. Women's long combined hours of paid and unpaid work have been documented for India and for other low- and middle-income countries (Hirway and Jose 2011; Antonopoulos and Hirway 2010; Charmes 2019), and time poverty has been measured in Guinea, Guatemala and parts of Latin America (Bardasi and Wodon 2010; Gammage 2010; Zacharias et al. 2012). Evidence on how the free-time gap in India varies over the life course, and on how much of it is a matter of life stage rather than social position, is scarce.

A single national average hides when in life the gap is incurred. A young unmarried woman, a mother of a toddler and a grandmother live very different days, and so do their male counterparts. If the gap is concentrated in a few family stages, policy that targets women by income or caste will reach the wrong people at the wrong time. This paper describes the free-time gap over the life course using all 533,719 diaries in India's Time Use Survey 2024 (National Statistics Office 2025). I define seven family stages from age, marital status and household composition. Free time is the sum of ICATUS 2016 divisions 7 and 8: socialising, community participation, religious practice, culture, leisure, mass media and sports (United Nations Statistics Division 2016).

The analysis has six parts. First, I estimate the gender gap in free time and in total work by family stage. Second, a Kitagawa decomposition separates the national gap into a within-stage component and a component due to men and women being distributed differently across stages (Kitagawa 1955). Third, I measure time poverty in three ways: a relative free-time line at 50% and 60% of the median, a long-hours line on total work, and a two-dimensional measure that crosses time poverty with consumption poverty (Burchardt 2008; Merz and Rathjen 2014). Fourth, I report every main result with and without religious practice in free time, because whether worship is discretionary is contested. Fifth, I compare the conditional dispersion of the gap across family stages with its dispersion across education, consumption, caste, religion and sector. Sixth, I examine two transitions that the life-course literature emphasises: retirement, by comparing pensioners with workers aged

55 to 70, and the journey to work, by measuring commuting among the employed.

The headline findings are as follows. Women have 15.1 minutes (SE 0.8) less free time than men, and do 29.9 minutes (SE 1.1) more total work. The gap is widest for persons "young, never married (15-24)" (-45.7 minutes (SE 1.7)) and narrowest for "married, no child under 15" (8.5 minutes (SE 1.4)). Of the national gap of -15.1 minutes, -11.6 minutes (bootstrap SE 0.8) arise within stages and -3.5 minutes (SE 0.2) from stage composition. The conditional gap is more dispersed across family stages than across any social dimension, so life-course position organises the gender gap in leisure more than social group does. Women are 2.1 percentage points (SE 0.2) more likely than men to be time-poor at the 60% line, and 2.6 percentage points (SE 0.2) more at the 50% line.

The paper contributes to the social-indicators literature in three ways. It provides a life-course profile of the free-time gap and of time poverty for India from a complete national diary survey. It shows how sensitive time-poverty comparisons are to the choice of line, to the treatment of religious practice and to whether income is considered jointly. And it consolidates into one framework a set of dimensions (parenthood, education, employment, retirement, commuting, religion, caste, household size) that are often studied one at a time.

2 Related literature

2.1 Measuring leisure and its distribution

Time-use research distinguishes three ways of defining leisure (Juster and Stafford 1991; Aguiar et al. 2012). The activity approach, used here, counts minutes spent in activities classified as leisure. The residual approach counts the time left over after paid work, unpaid work and personal care. The subjective approach asks people whether they experienced an activity as free. The activity approach has the advantage of being replicable across surveys that share a classification such as ICATUS, and it is the basis for the long-run series in Robinson and Godbey (1997), Gershuny (2000), Aguiar and Hurst (2007) and Ramey and Francis (2009). Aguiar and Hurst (2007) show that leisure in the United States rose between 1965 and 2003, and that the gains were largest for less-educated adults. Sevilla et al. (2012) find that leisure inequality by education grew over the same period in both quantity and quality. Cross-country comparisons confirm rising leisure in most industrialised countries (Gimenez-Nadal and Sevilla 2012).

Minutes of leisure are not the same as well-being. The Day Reconstruction Method showed that activities differ widely in experienced affect, so the composition of free time matters as well as its quantity (Kahneman et al. 2004; Kahneman and Krueger 2006). National time accounting proposes weighting time by its experienced quality (Krueger et al. 2009). Goodin et al. (2008) go further and define discretionary time: the hours left after the time strictly necessary to reach a socially defined minimum in income, household work and personal care. Goodin et al. (2005) show that free time and discretionary time can give different answers about who is pressed for time. Other work stresses that the experience of time pressure has a social meaning of its own (Gershuny 2005; Wajcman

2015) and that self-reported time stress rises with earnings and family obligations (Hamermesh and Lee 2007). Hamermesh (2019) surveys this economics of time.

2.2 Gender and the quantity and quality of free time

The gender gap in free time is one of the most stable findings in time-use research. In the United States, women have less free time than men, and their free time is more often combined with other activities, especially childcare (Mattingly and Bianchi 2003). In Australia, women's leisure is more fragmented and less often "pure" adult leisure (Bittman and Wajcman 2000). Sayer (2005) shows that the gender gap in free time persisted while men's and women's paid and unpaid work converged. Leisure quality matters for the gender comparison because overlapping activities intensify work (Floro 1995; Floro and Miles 2003). Mattingly and Sayer (2006) find that the relation between free time and feeling rushed differs between women and men.

The arrival of children is the life-course event most clearly associated with lost leisure. Mothers absorb more of the extra childcare time than fathers (Bianchi et al. 2006; Craig 2007; Craig and Mullan 2011), and in a comparison of five countries mothers have less leisure than fathers, with less of it free of children (Craig and Mullan 2013). Gender differences in free time are also shaped by class: among British workers, free-time poverty is concentrated in particular class positions as well as among women (Chatzitheochari and Arber 2012). Whether the gender gap in free time is larger across family stages or across social groups is therefore an empirical question.

2.3 Time poverty: competing measures

Vickery (1977) introduced time poverty by noting that a household needs a minimum of both money and time for home production, so that an income poverty line that ignores time understates deprivation among households with few adults. Three families of measures have since developed (Williams et al. 2016).

The first sets a threshold on free time, typically a fraction of the median, in the tradition of relative income poverty lines such as the European convention of 60% of median income (Atkinson et al. 2002). Relative free-time lines have been used to compare groups within countries (Chatzitheochari and Arber 2012), and Kalenkoski et al. (2011) derive time-poverty thresholds and rates for the US population that differ by household type. The second sets a threshold on working time. Bardasi and Wodon (2010) classify as time-poor those whose total paid and unpaid working hours exceed a multiple of the median, and show that women in Guinea are more often time-poor than men. Blackden and Wodon (2006) and Gammage (2010) document long working hours and their gendered distribution in Sub-Saharan Africa and Guatemala. The third family is two-dimensional. Burchardt (2008) models the joint space of time and income for British households; Harvey and Mukhopadhyay (2007) measure the time poverty of working parents in Canada; Merz and Rathjen (2014) propose an interdependent measure for Germany in which a shortfall in one dimension can be compensated by the other; and Zacharias et al. (2012) construct a time-adjusted income poverty measure for

Latin American countries in which the cost of replacing missing household production is added to the poverty line. Multidimensional counting approaches give a general framework for such joint measures (Alkire and Foster 2011), and poverty gap indices summarise depth as well as incidence (Foster et al. 1984).

The choice among these measures matters. A free-time line classifies as poor people with long hours of personal care; a work-time line classifies as poor anyone with long hours, including the well paid. Two-dimensional measures deal with the second objection but require income data. This paper reports all three, with consumption standing in for income.

2.4 The life course and retirement

The life-course perspective treats the timing of transitions, and the lives linked to one's own, as determinants of outcomes at each age (Elder 1994). For time use, the transitions that matter most are marriage, the arrival and ageing of children, and exit from paid work. Free time rises at older ages in high-income countries, with much of the gain going to passive leisure such as television (Gauthier and Smeeding 2003). Retirement frees time but part of it goes to home production: Aguiar and Hurst (2005) show that retirees spend more time on food production and shopping, and Stancanelli and Van Soest (2012) find, using a regression discontinuity at the French retirement age, that retirement raises housework. Whether retirement narrows or widens the gender gap in free time therefore depends on how men and women reallocate the freed hours. In India, formal retirement with a pension is the experience of a minority of older people, and population ageing is raising the policy salience of their well-being (International Institute for Population Sciences et al. 2020; UNFPA and International Institute for Population Sciences 2023).

2.5 Commuting and well-being

Commuting is a large and growing claim on the working day in urbanising economies, and it is among the least enjoyed activities of the day (Kahneman et al. 2004). Stutzer and Frey (2008) find that German commuters report lower life satisfaction and are not compensated by higher wages or better housing, a "commuting paradox". Roberts et al. (2011) find that commuting harms women's psychological health more than men's in Britain, which they attribute to women's greater responsibility for household work. Women tend to work closer to home (Hanson and Pratt 1995) and, in French job-search data, value shorter commutes more than men do (Le Barbanchon et al. 2021). A shorter commute leaves more time for other activities but may come with a lower wage or a narrower set of jobs.

2.6 Leisure and time use in developing countries and India

Time-use surveys in the Global South have expanded rapidly, and they show that measuring women's work requires capturing unpaid and own-use production (Esquivel et al. 2008; Antonopoulos and Hirway 2010). In India, analyses of the 1998–99 pilot time-use survey documented women's large

unpaid workloads (Hirway and Jose 2011), and the first national survey in 2019 provided national estimates (National Statistical Office 2020). Women's paid work in India is shaped by marriage, social norms about mobility and household status (Klasen and Pieters 2015; Afridi et al. 2018; Jayachandran 2021; Fletcher et al. 2017), and these same constraints bear on how women can use free time outside the home. Total work tends to be higher for women than for men in poorer countries (Burda et al. 2013), so in India the free-time gap is likely to reflect a total-work gap rather than a different mix of paid and unpaid work. The life-course profile of free time in India, and the robustness of gendered time-poverty comparisons to measurement choices, have not been documented from a complete national diary survey.

3 Data and measures

3.1 The Time Use Survey 2024

The Time Use Survey 2024 was fielded by the National Statistics Office between January and December 2024 (National Statistics Office 2025). Members aged six and above in each sampled household reported their activities over the 24 hours from 4 a.m. on the day before the interview, in 30-minute slots following the design of the 2019 round, with each activity coded to the ICATUS 2016 three-digit classification (United Nations Statistics Division 2016). Each slot can hold several activities, one of which is flagged as the major activity. I use every activity record (10,211,478 records, 533,719 diaries, 139,487 households in 9,969 first-stage units). Major activities are summed to the person-day, so each diary sums to 1,440 minutes (100.0% of diaries do). Minor activity records are used only to measure secondary activities during free time. Table 8 in the appendix summarises the construction checks.

The analysis sample is the 388,847 men and women aged 15 and above. Every person can be assigned to a family stage, so the sample is 388,847 persons. All estimates use the survey multipliers.

3.2 Free time and its components

Free time is the sum of minutes in ICATUS divisions 7 (socialising and communication, community participation, religious practice) and 8 (culture, leisure, mass media and sports). I report ten components: socialising and talking, community participation, religious practice, other social activities, attending cultural events, hobbies and games, exercise and sport, mass media, resting and relaxing, and other leisure. Because religious practice is an obligation for some and a chosen activity for others, I construct a second measure, free time excluding religious practice, and report every main result for both.

Total work is the sum of employment (including employment-related travel and commuting), own-use production of goods, unpaid domestic services, unpaid care for household members, and unpaid volunteer work: ICATUS divisions 1 to 5. Commuting is ICATUS code 182, travel between home and the place of work.

Two measures of free-time quality are available. Secondary free time is minutes of free-time activities recorded as minor activities in a slot. Free time accompanied by unpaid work is the number of minutes of minor domestic or care activities recorded within a major free-time episode, matched by start time. This is a lower bound on what Mattingly and Bianchi (2003) call contaminated free time, because only activities that respondents report as simultaneous are captured.

3.3 Family stages

I assign each person to one of seven stages:

1. *Young, never married (15–24).*
2. *Married, child under 6 in household*: currently married, below 60, and at least one child under six lives in the household.
3. *Married, child 6–14 in household*: as above, with no child under six and at least one child aged 6 to 14.
4. *Married, no child under 15*: currently married, below 60, with no child under 15 in the household.
5. *Other not married (15–59)*: never married aged 25 to 59, and widowed, divorced or separated persons aged 15 to 59.
6. *Aged 60–69*, and 7. *Aged 70+*, regardless of marital status.

Children under six do not keep diaries, so their number is household size minus the number of diarists. Children are counted at the household level, so the "child under 6" stage includes grandparents below 60 living with young grandchildren. The stages are cross-sectional positions, not transitions observed for the same people; the profile they trace combines life-course and cohort differences.

3.4 Poverty lines

The relative time-poverty lines are 50% and 60% of the weighted median of free time among persons 15 and above (210 minutes, so the lines are 105 and 126 minutes). At the 60% line I also report the poverty gap index, the mean of the proportional shortfall below the line (Foster et al. 1984). The line for free time excluding religious practice is 60% of its own median (210 minutes). The long-hours line follows the logic of Bardasi and Wodon (2010): total work above 1.5 times its median (480 minutes). For the two-dimensional measure, a person is consumption-poor if household monthly per capita consumption expenditure (MPCE) is below 60% of the person-weighted median (2,842 rupees). Crossing this with the 60% free-time line gives four groups: neither poor, time-poor only, consumption-poor only, and both. Consumption is measured for the household, so any gender gap in consumption poverty reflects only differences in the households that men and women live in, not intra-household allocation.

4 Methods

All means and gaps are survey-weighted. Gaps are reported as women minus men, so a negative free-time gap means women have less. Standard errors are clustered on the first-stage unit, the primary sampling unit of the survey design (Cameron and Miller 2015; Abadie et al. 2023). Strata are not used, which makes the standard errors conservative.

Decomposition. Let s_{gk} be the population share of sex g in stage k and $\bar{F}_{gk}$ the mean free time of that cell. The national gap $\bar{F}_W - \bar{F}_M = \sum_k s_{Wk}\bar{F}_{Wk} - \sum_k s_{Mk}\bar{F}_{Mk}$ is decomposed following Kitagawa (1955) as

$$\bar{F}_W - \bar{F}_M = \underbrace{\sum_k \tfrac{s_{Wk}+s_{Mk}}{2}\,(\bar{F}_{Wk} - \bar{F}_{Mk})}_{\text{within stages}} + \underbrace{\sum_k (s_{Wk} - s_{Mk})\,\tfrac{\bar{F}_{Wk}+\bar{F}_{Mk}}{2}}_{\text{stage composition}}.$$

The within-stage component is the gap that would remain if men and women were distributed identically across stages. The composition component arises because, for example, women marry younger than men and are more often widowed. The symmetric weights avoid the index-number problem of choosing men or women as the reference, which arises in the regression-based analogue (Oaxaca 1973; Blinder 1973). Standard errors for the two components come from 199 bootstrap replications that resample first-stage units.

Heterogeneity. For each dimension D (family stage, education, consumption quintile, social group, religion, sector) I estimate

$$F_i = \sum_d \gamma_d \,\mathbf{1}[D_i = d] \times W_i + \sum_d \alpha_d \,\mathbf{1}[D_i = d] + X_i'\beta + \mu_{s(i)} + \tau_{t(i)} + \varepsilon_i,$$

where W_i indicates a woman, X_i contains the other dimensions, household size and an indicator for a proxy-reported diary, and μ and τ are state and day-of-week fixed effects. Each γ_d is the conditional women-minus-men gap at level d. I summarise how much the gap varies along a dimension by the population-weighted standard deviation of the γ_d across its levels, and also report the range. The comparison of dispersions is descriptive: a dimension with more levels or finer categories can show more dispersion mechanically.

Retirement. Among persons aged 55 to 70 whose usual principal status is either working or pensioner, I regress each time-use outcome on a pensioner indicator, single-year-of-age dummies, sector, education, consumption quintile, social group, religion, household size, proxy report, and state and day fixed effects, separately for men and women. A pooled model with a pensioner-by-woman interaction tests whether the pensioner difference differs by sex. Status is self-reported and retirement is not randomly assigned, so the estimates describe how pensioners' days differ from those of otherwise similar workers, not the causal effect of retirement.

Commuting. Among persons whose usual status is working and who did some employment on the

diary day, I estimate the gender gap in commuting and the association between commuting minutes and free time, holding minutes of paid work, age band, sector, education, consumption quintile, family stage, social group and household size constant.

5 Results

5.1 Free time and total work over the life course

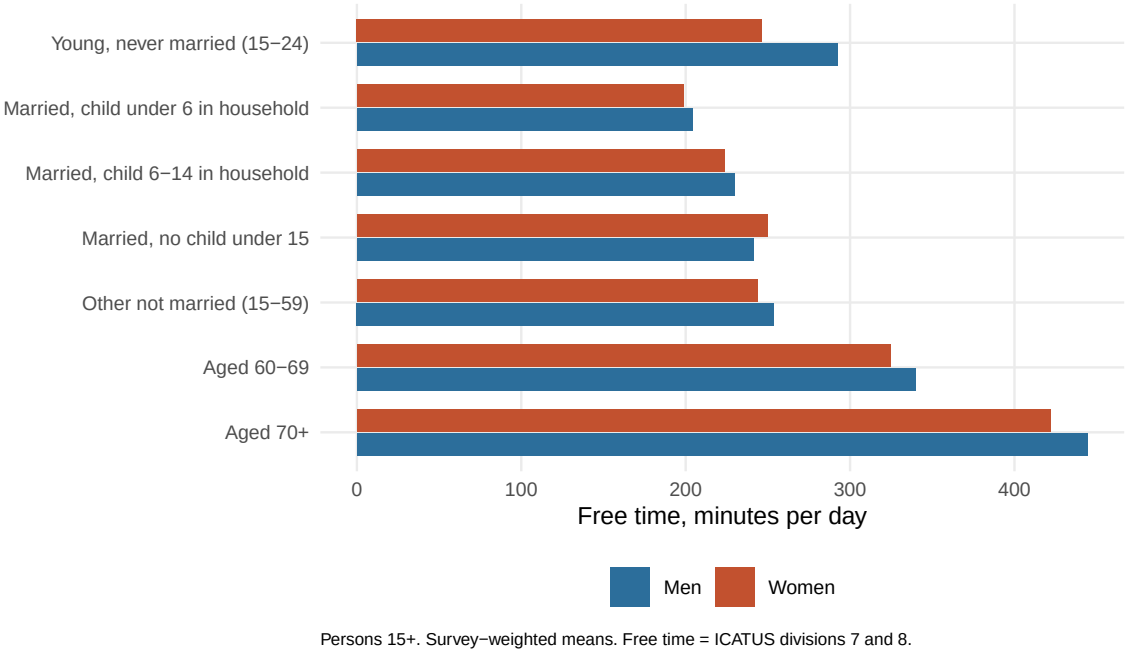


Figure 1: Free time by family stage and sex

Table 1: Women-minus-men gaps in free time and total work by family stage, minutes per day

Stage	N men	N women	Free time	Free time excl. religion	Total work
Young, never married (15-24)	37,434	27,243	-45.7*** (1.7)	-48.5*** (1.7)	-12.2*** (2.5)
Married, child under 6 in household	51,736	59,107	-5.5*** (1.1)	-11.7*** (1.1)	18.0*** (1.4)
Married, child 6-14 in household	20,429	22,898	-6.1*** (1.6)	-14.5*** (1.5)	29.2*** (1.9)
Married, no child under 15	36,722	41,982	8.5*** (1.4)	-2.7** (1.3)	2.0 (1.8)
Other not married (15-59)	19,558	15,765	-9.9*** (2.3)	-20.5*** (2.2)	13.6*** (3.1)
Aged 60-69	18,317	18,236	-15.2*** (2.1)	-27.0*** (2.0)	14.7*** (2.6)
Aged 70+	9,709	9,711	-22.4*** (2.7)	-34.4*** (2.6)	10.7*** (3.2)

Notes: Persons 15+. Survey-weighted differences in means, women minus men; FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Figure 1 shows mean free time by stage and sex, and Table 1 reports the women-minus-men gaps. Men aged 15 and above have 261 minutes of free time a day and women 246. The gap is not uniformly negative and significant across stages. The gap is widest for "young, never married (15-24)" and narrowest for "married, no child under 15"; the difference between these two stages is

54 minutes.

Women do significantly more total work than men at most stages, so the free-time gap reflects a gap in total work and not merely a different mix of paid and unpaid work. This is the pattern expected for a lower-income country (Burda et al. 2013). Across all persons 15+, women's sleep differs from men's by 11.1 minutes (SE 0.5), all personal care by -7.6 minutes (SE 0.5) and learning by -7.2 minutes (SE 0.6). Because the day is fixed at 1,440 minutes, the free-time gap equals minus the sum of the gaps in total work, learning and personal care; these figures show where women's shortfall in free time is absorbed.

5.2 Within-stage gaps and stage composition

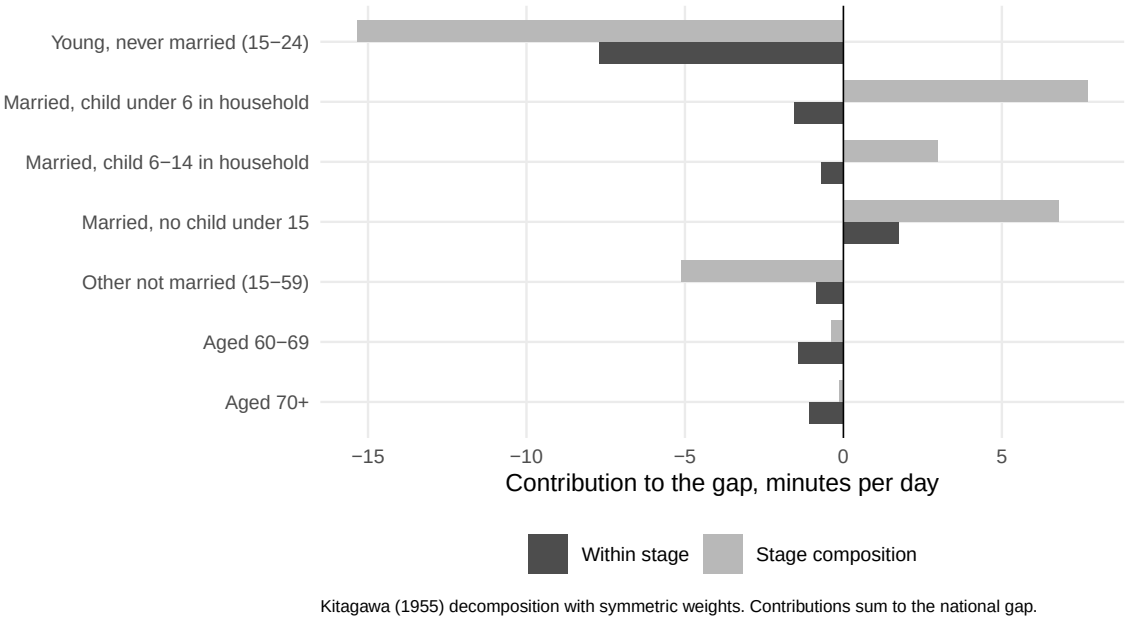


Figure 2: Contributions of each family stage to the women-minus-men free-time gap

Table 2: Decomposition of the women-minus-men free-time gap by family stage

Stage	Share of men	Share of women	Free time, men	Free time, women	Within stage	Composition
Young, never married (15-24)	19.7%	14.0%	292.2	246.5	-7.7	-15.4
Married, child under 6 in household	26.3%	30.1%	204.3	198.8	-1.5	7.7
Married, child 6-14 in household	10.7%	12.0%	229.7	223.6	-0.7	3.0
Married, no child under 15	19.2%	22.0%	241.2	249.8	1.8	6.8
Other not married (15-59)	9.6%	7.6%	253.8	243.9	-0.9	-5.1
Aged 60-69	9.5%	9.4%	340.0	324.8	-1.4	-0.4
Aged 70+	4.9%	4.9%	444.5	422.1	-1.1	-0.1
Total	100%	100%	260.9	245.8	-11.6 (0.8)	-3.5 (0.2)

Notes: Persons 15+. Survey-weighted. Within-stage and composition contributions use symmetric (average) weights and sum to the national gap of -15.1 minutes. Bootstrap SEs (199 replications resampling FSUs) for the totals in parentheses. Excluding religious practice, the within-stage component is -19.7 (0.7) and the composition component -3.9 (0.1).

Table 2 and Figure 2 decompose the national gap of -15.1 minutes. The within-stage component is -11.6 minutes (SE 0.8) and the stage-composition component is -3.5 minutes (SE 0.2). Most of the

national gap therefore arises because women have less free time than men at the same family stage, not because women are concentrated in stages where everyone has little free time. Composition accounts for 23% of the gap and is statistically distinguishable from zero. Composition widens the gap: women are over-represented in stages where free time is scarce. Without religious practice, the within-stage component is -19.7 minutes (SE 0.7) and the composition component -3.9 minutes (SE 0.1).

The composition component reflects demography. Women in India marry younger than men (Desai and Andrist 2010), so more women than men of a given age are in the married stages, and widowhood is more common among women: 12.2% of women aged 15 and above are widowed, against 3.6% of men. A national average that ignores this mixes a gap between equals with a gap that arises from where men and women are in the life cycle.

5.3 Time poverty: sensitivity to the line and the second dimension

Table 3: Time poverty and joint time-consumption poverty by sex

Measure	Men	Women	Women minus men (pp)
Free time below 50% of median	15.4%	18.0%	2.6*** (0.2)
Free time below 60% of median	23.6%	25.7%	2.1*** (0.2)
Free-time poverty gap index, 60% line (x 100)	8.3	10.2	1.9*** (0.1)
Free time excl. religious practice below 60% of its median	25.8%	29.7%	3.9*** (0.3)
Total work above 1.5 times the median	3.7%	6.7%	3.0*** (0.1)
Consumption-poor (MPCE below 60% of median)	12.6%	13.6%	1.0*** (0.1)
Time-poor and consumption-poor	3.0%	4.1%	1.1*** (0.1)
Time-poor only	20.5%	21.6%	1.0*** (0.2)
Consumption-poor only	9.6%	9.5%	-0.0 (0.1)

Notes: Persons 15+. Survey-weighted. Median free time 210 minutes; median free time excluding religious practice 210; median total work 480; person-weighted median MPCE 2,842 rupees. The last three rows use the 60% free-time line and partition the population with the 'neither poor' group. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 3 reports time poverty under each definition. Women are 2.1 percentage points (SE 0.2) more likely than men to have free time below 60% of the median and 2.6 percentage points (SE 0.2) more likely at the 50% line. The sign and significance of the gender gap in time poverty do not depend on the choice between the two lines. The poverty gap index, which counts how far below the line the poor fall, differs by 1.9 points (SE 0.1), women minus men. Excluding religious practice from free time, the gap in the headcount is 3.9 percentage points (SE 0.3) higher for women. On the long-hours definition, women are 3.0 percentage points (SE 0.1) more likely than men to be time-poor. The free-time and work-time definitions need not agree, because the first is affected by time in personal care and learning and the second is not.

The two-dimensional measure shows how time poverty and consumption poverty overlap. The share who are both time-poor and consumption-poor is 1.1 percentage points (SE 0.1) higher for women than for men, and the gap in consumption poverty alone is 1.0 percentage points (SE 0.1) higher

for women. Among women, the time-poverty rate of the consumption-poor differs from that of the non-poor by 5.2 percentage points (SE 0.6) higher; among men by 0.5 percentage points (SE 0.5) higher. Time poverty among women is concentrated in consumption-poor households, so the two deprivations compound.

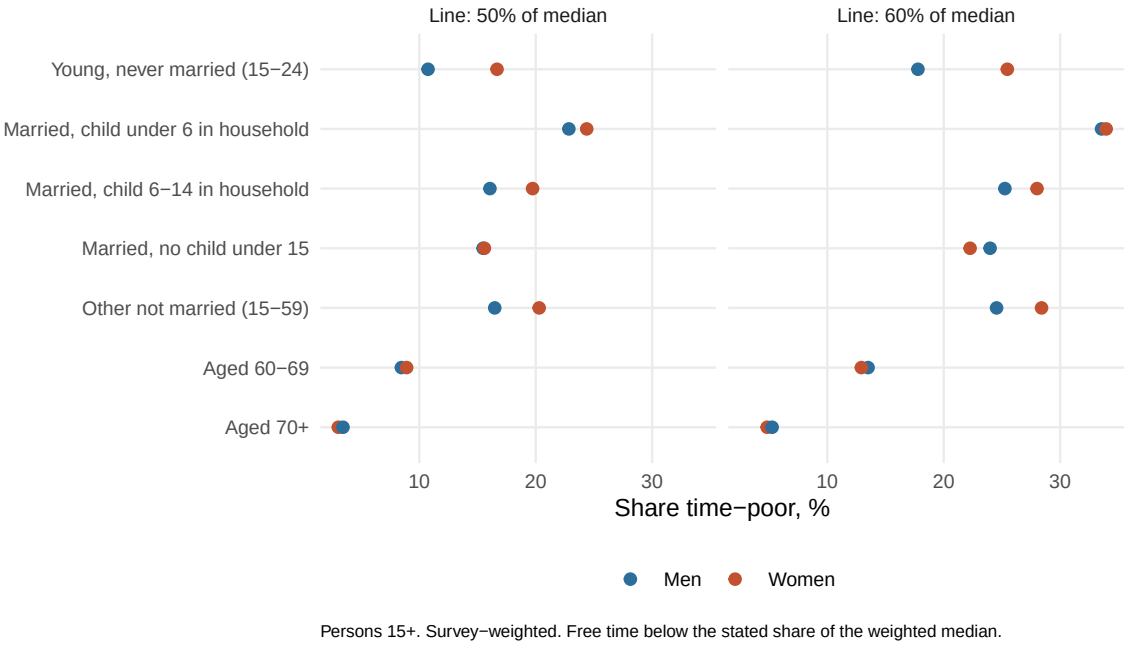


Figure 3: Time-poverty headcount by family stage, sex and poverty line

Figure 3 shows time poverty by stage at both lines. The gender gap in the 60% headcount is largest for "young, never married (15-24)" and, at the 50% line, for "young, never married (15-24)". The stage at which women are most disadvantaged is the same under both lines. The rank correlation of the stage-specific gaps across the two lines is 0.89.

5.4 Religious practice

Women spend 20 and men 11 minutes a day on religious practice, a gap of 8.5 minutes (SE 0.2). Excluding it, the women-minus-men gap in free time is -23.6 minutes (SE 0.8), against -15.1 minutes (SE 0.8) with it. Because women spend significantly more time than men on religious practice, counting it as free time narrows the measured gap; on the narrower definition the gap is wider. Across stages, the rank correlation between the gaps with and without religious practice is 1.00 (Table 1), and the stage with the widest gap is the same on both definitions. Readers who regard worship as obligatory rather than discretionary can use the narrower measure throughout; the decomposition and poverty results are reported for both.

Table 4: Dispersion of the women-minus-men free-time gap across levels of each dimension, minutes

Dimension	Levels	Range, raw	SD, raw	Range, conditional	SD, conditional
Family stage	7	54.3	17.4	57.8	18.3
Education	4	19.4	6.6	16.2	5.5
Consumption quintile	5	23.1	8.9	21.3	7.9
Social group	4	41.1	10.9	41.6	11.0
Religion	5	44.9	6.1	46.8	6.4
Sector	2	39.6	18.4	35.2	16.4

Notes: Raw: women-minus-men gap estimated separately within each level. Conditional: coefficients on level-by-woman interactions in a model with all other dimensions, household size, proxy report, and state and day-of-week fixed effects. SD is the population-weighted standard deviation of the level-specific gaps. Level-specific estimates are in the appendix. Baseline conditional gap (common across levels): -12.5 minutes (SE 0.8).

5.5 Does social position matter as much as life stage?

Table 4 compares how much the gap varies along each dimension; the level-specific gaps are in Table 9 in the appendix. The pooled conditional gap is -12.5 minutes (SE 0.8). Across family stages the weighted standard deviation of the conditional gap is 18.3 minutes; the largest standard deviation for any other dimension is 16.4 minutes, for sector. Conditional on the other dimensions, the gap is more dispersed across family stages than across education, consumption, caste, religion or sector. The unconditional comparison gives a different ordering, so part of the apparent variation along social dimensions is due to their correlation with family stage, or vice versa. Because consumption quintiles and education tiers are ordered, Table 9 also shows whether the gap rises or falls along them.

5.6 What free time consists of and its quality

Table 5: Composition of free time by sex, minutes per day and participation

Component	Men	Women	Women minus men	Doing any, men	Doing any, women
Socialising and talking	84.9	82.5	-2.4*** (0.4)	74%	74%
Community participation	2.9	2.5	-0.4*** (0.1)	2%	1%
Religious practice	11.5	20.0	8.5*** (0.2)	17%	34%
Other social activities	4.9	3.4	-1.5*** (0.1)	5%	3%
Attending cultural events	1.0	0.6	-0.4*** (0.0)	1%	1%
Hobbies and games	13.1	3.5	-9.6*** (0.2)	10%	4%
Exercise and sport	10.5	2.8	-7.7*** (0.1)	17%	5%
Mass media (TV, radio, reading)	86.1	76.1	-10.0*** (0.4)	72%	60%
Resting and relaxing	44.4	53.5	9.2*** (0.3)	52%	56%
Other leisure	1.6	0.7	-0.9*** (0.1)	2%	1%

Notes: Persons 15+. Survey-weighted means over all persons (including non-participants). Participation is the share with any minutes of the activity as a major activity on the diary day. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Table 5 breaks free time into its components. The component in which women's shortfall is largest is mass media (tv, radio, reading). Women differ from men by -10.0 minutes (SE 0.4) in mass media,

-2.4 minutes (SE 0.4) in socialising and talking, and 9.2 minutes (SE 0.3) in resting. Women spend significantly more time than men in religious practice, resting and relaxing.

Two quality measures point in the direction of the literature on contaminated leisure (Mattingly and Bianchi 2003; Bittman and Wajcman 2000). Secondary free time, recorded alongside another major activity, differs by -17.6 minutes (SE 0.7), women minus men. Minutes of secondary domestic or care work recorded during free-time episodes differ by 5.7 minutes (SE 0.1), women minus men. These measures capture only what respondents recorded as simultaneous, so they understate the extent to which free time is interrupted or supervised.

5.7 Retirement: pensioners and workers aged 55 to 70

Table 6: Pensioners minus workers aged 55 to 70, conditional differences in minutes per day

Outcome	Men	Women	Difference, women minus men
Free time	255.1*** (2.9)	189.1*** (3.9)	-62.0*** (3.9)
Free time excl. religious practice	243.1*** (2.9)	174.4*** (3.8)	-65.7*** (3.8)
Religious practice	11.9*** (0.9)	14.7*** (1.1)	3.7*** (1.3)
Socialising and talking	78.1*** (2.1)	69.6*** (2.5)	-4.2 (2.6)
Mass media	93.3*** (2.1)	58.5*** (2.3)	-36.4*** (2.6)
Employment	-374.3*** (2.8)	-282.5*** (3.6)	96.4*** (3.3)
Unpaid domestic and care work	35.0*** (1.6)	3.4 (3.5)	-40.3*** (3.4)
Total work	-347.5*** (2.9)	-285.9*** (4.3)	55.3*** (4.1)
Sleep	85.3*** (1.8)	86.8*** (2.4)	5.8** (2.4)

Notes: Persons aged 55-70 whose usual principal status is working or 'rentier, pensioner, remittance recipient'. N: men 23,686 workers and 6,461 pensioners; women 8,343 workers and 4,375 pensioners. Controls: single years of age, urban, education, consumption quintile, social group, religion, household size, proxy report; state and day-of-week fixed effects. The last column is the pensioner-by-woman interaction in a pooled model with common controls. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

Table 6 compares the days of pensioners and workers aged 55 to 70. Male pensioners have 255.1 minutes (SE 2.9) more free time than comparable male workers, and female pensioners 189.1 minutes (SE 3.9) more than comparable female workers. Part of the time freed from paid work goes to unpaid domestic and care work, consistent with the reallocation to home production found in high-income countries (Aguiar and Hurst 2005; Stancanelli and Van Soest 2012). Unpaid domestic and care work differs between pensioners and workers by 35.0 minutes (SE 1.6) for men and 3.4 minutes (SE 3.5) for women. The pensioner difference in free time differs significantly between men and women. Among pensioners the women-minus-men free-time gap is -119.7 minutes (SE 3.5), against -58.2 minutes (SE 2.1) among workers of the same ages, so the gap is not narrower among pensioners.

These comparisons describe the pensioner population, which is likely to be selected toward formal and urban employment, and the "pensioner" status also covers rentiers and remittance recipients. The estimates should not be read as the effect of retiring.

Table 7: Commuting and time use among employed persons

Outcome	Men	Women	Gap, all	Gap, urban	Gap, rural
Any commuting on diary day (share)	76.7%	62.8%	-13.9 (0.5)	-15.5 (0.8)	-13.0 (0.6)
Commuting, minutes	61.8	45.0	-16.8 (0.4)	-16.6 (0.7)	-16.6 (0.5)
Paid work excl. travel, minutes	430.3	337.6	-92.7 (1.3)	-96.6 (2.0)	-88.1 (1.5)
Commuting as share of employment time	12.5%	10.8%	-1.7 (0.1)	-1.7 (0.1)	-1.7 (0.1)
Free time, minutes	198.5	140.1	-58.4 (1.0)	-33.5 (1.6)	-70.2 (1.2)
Total work, minutes	544.0	635.5	91.5 (1.2)	60.8 (2.2)	106.4 (1.5)

Notes: Persons 15+ whose usual principal status is working and who report some employment on the diary day (N = 172,987). Gaps are women minus men; shares are in percentage points. Commuting is ICATUS code 182. Survey-weighted; FSU-clustered SEs in parentheses.

5.8 Commuting among the employed

Table 7 describes commuting among the 172,987 employed persons who worked on the diary day. Employed women commute 16.8 minutes (SE 0.4) less than employed men (16.6 minutes (SE 0.7) less in urban areas and 16.6 minutes (SE 0.5) less in rural areas). Women are 13.9 percentage points (SE 0.5) less likely than men to commute at all. Among those who commute, the gap is -8.8 minutes (SE 0.3). Women's shorter commutes are consistent with evidence that women work closer to home (Hanson and Pratt 1995; Le Barbanchon et al. 2021), which in India may also reflect constraints on women's mobility (Jayachandran 2021).

Holding paid-work minutes and other characteristics constant, each additional minute of commuting is associated with -0.54 minutes (SE 0.01) of free time for men and -0.29 minutes (SE 0.02) for women. The difference between women and men is 0.41 minutes (SE 0.02). There is no evidence that commuting displaces more of women's free time than of men's. Among workers overall, women have 58.4 minutes (SE 1.0) less free time than men. These associations are descriptive: commute length, hours and free time are chosen jointly, and the diary records a single day.

6 Discussion

Indian women have less free time than men, and, conditional on covariates, the gap varies more with family stage than with education, consumption, caste, religion or sector. Four features of the results bear on how free time and time poverty should be measured and used as social indicators.

First, the national average is a mixture. The decomposition separates a within-stage gap from a gap that arises because women and men occupy different stages. The within-stage component dominates, so most of the gap is between men and women at the same point in the life cycle. Reporting the gap by family stage, alongside the national figure, would make SDG 5.4 reporting on unpaid work and its leisure counterpart easier to interpret and to compare across countries with different demographic structures.

Second, time-poverty comparisons are sensitive to the definition. The free-time and long-hours

definitions answer different questions, the line can move the location of the peak across stages, and the treatment of religious practice changes measured free time by 20 minutes for women and 11 minutes for men. I recommend that studies of time poverty in South Asia report at least two lines and both treatments of religious practice, as this paper does. The two-dimensional measure shows that time poverty and consumption poverty are distinct; because they overlap among women, programmes that add to the working hours of poor women, such as public works, should count the time cost as well as the income gain.

Third, the life-course pattern has a direct policy reading. Interventions that target women by income or caste alone will miss the main source of variation in women's free time, which is the stage of the family life cycle. The stages with the widest gap are natural targets for time-saving services, such as childcare, piped water and clean cooking fuel. Their effect on women's free time, rather than on paid work alone, deserves evaluation.

Fourth, retirement and commuting show how transitions in and around paid work redistribute free time. Among people in their late fifties and sixties, the gender gap in free time is not narrower among pensioners than among workers, and employed women commute less than employed men. Both patterns describe a selected population: pensions and long commutes are features of formal and urban employment, which few Indian women hold.

7 Limitations

The design is a single cross-section. Stages are positions occupied by different people, not transitions experienced by the same people, so cohort differences are confounded with life-course differences. The single diary day records one day per person; it estimates group means without bias but overstates the dispersion of individual time use, which matters for headcount measures of time poverty (Frazis and Stewart 2012). Free time is defined by activity, so it includes activities that some respondents experience as obligations and excludes leisure taken while doing something else. No measure of experienced well-being is available, so the paper cannot say whether the minutes women lose are minutes they would have enjoyed (Kahneman et al. 2004). Consumption is measured for the household, so consumption poverty does not capture intra-household inequality, and the two-dimensional measure does not adjust the consumption line for the cost of replacing household production, as Zacharias et al. (2012) do. Children under six are counted from household size, and only at the household level. Finally, the dispersion comparison across dimensions depends on how finely each dimension is divided.

8 Conclusion

Using every diary in India's Time Use Survey 2024, this paper has traced the gender gap in free time across seven family stages, decomposed the national gap into within-stage and composition components, measured time poverty with three definitions and two lines, reported every main result

with and without religious practice, and examined two transitions around paid work, retirement and commuting. Women have less free time than men, and the size of their shortfall depends more on where they are in the family life cycle than on their education, consumption, caste or religion. Free time and time poverty should be reported by life stage, with the definition and line stated, if they are to serve as indicators of gender equality in well-being.

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Appendix

Table 8: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	533,719
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.

Table 9: Women-minus-men free-time gap by level of each dimension, minutes per day

Dimension	Level	N	Raw gap	Conditional gap
Family stage	Young, never married (15-24)	64,677	-45.7 (1.7)	-49.9 (1.7)
	Married, child under 6 in household	110,843	-5.5 (1.1)	-8.1 (1.1)
	Married, child 6-14 in household	43,327	-6.1 (1.6)	-7.9 (1.6)
	Married, no child under 15	78,704	8.5 (1.4)	8.0 (1.3)
	Other not married (15-59)	35,323	-9.9 (2.3)	-5.0 (2.2)
	Aged 60-69	36,553	-15.2 (2.1)	-11.0 (2.1)
	Aged 70+	19,420	-22.4 (2.7)	-17.2 (2.7)
Education	Not literate / below primary	95,772	-9.5 (1.4)	-6.5 (1.3)
	Primary / middle	110,809	-14.9 (1.3)	-9.3 (1.2)
	Secondary / higher secondary	123,473	-22.1 (1.2)	-15.7 (1.2)
	Graduate and above	58,793	-28.9 (1.7)	-22.7 (1.6)
Consumption quintile	Q1	61,242	-22.9 (1.6)	-20.8 (1.6)
	Q2	69,528	-22.9 (1.5)	-19.5 (1.4)
	Q3	77,774	-18.5 (1.4)	-15.2 (1.4)
	Q4	85,093	-14.0 (1.4)	-11.4 (1.3)
	Q5	95,210	0.2 (1.5)	0.6 (1.4)
Social group	Others	104,134	-5.2 (1.4)	-3.0 (1.4)
	OBC	166,497	-14.4 (1.1)	-11.6 (1.0)
	SC	66,627	-13.8 (1.6)	-11.7 (1.6)
	ST	51,589	-46.3 (2.2)	-44.5 (2.2)
Religion	Hindu	305,398	-16.2 (0.9)	-13.3 (0.9)
	Muslim	47,471	-5.1 (2.1)	-2.2 (2.1)
	Christian	21,746	-36.8 (3.9)	-38.5 (3.7)
	Sikh	6,424	8.2 (4.9)	8.3 (4.9)
	Other	7,808	-32.9 (5.8)	-28.7 (5.8)
Sector	Rural	241,050	-27.5 (1.0)	-23.7 (1.0)
	Urban	147,797	12.0 (1.3)	11.4 (1.3)

Notes: Raw: survey-weighted difference in means within the level. Conditional: coefficient on the level-by-woman interaction in a model that also includes the level main effects, the other dimensions, household size, proxy report, and state and day-of-week fixed effects. FSU-clustered SEs in parentheses.